

This file is only needed for 3D lattices to visualize epsilon in 3D

[Reference](#)

This can only be used if and only if all the steps in the "Installation.md" are followed as such, and all packages are installed without errors

After computation of band structure of the 3D lattice, say the file is at

```
~/Desktop/FCC
```

We have "band-epsilon.h5" in "Desktop/FCC"; it also has "band-dpwr.k.b.h5", say "band-dpwr.k06.b*.h5"

Refer to mpb-docs for more details on the commands

Now open a terminal at the location and run (change the numbers "06")

```
cd ~/Desktop/FCC
```

```
mpb-data -m 2 -r -n 32 band-epsilon.h5 band-dpwr.k06.b*.h5
```

```
h5tov5d -o fcc.v5d -d data-new band-epsilon.h5 band-dpwr.k06.b*.h5
```

Sometimes the HDF5 library is not found; then export the lib location (Replace "HDF5_PATH" with the HDF5 directory)

```
export LD_LIBRARY_PATH=HDF5_PATH/lib/:$LD_LIBRARY_PATH
```

If an HDF5 error pops up, export the following and run the above

```
export HDF5_USE_FILE_LOCKING=FALSE
```

This command may be used constantly in dealing with HDF5 files

To view using vis5d (Recommended)

```
vis5d fcc.v5d &
```

To view in Mayavi, execute

```
python plot3d.py
```